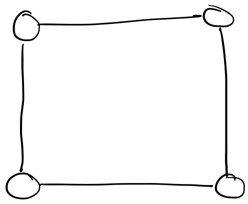
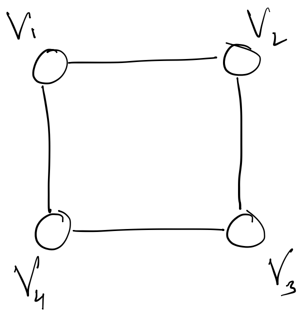
Lecture 15 — 14th October

Graph : A set of vertices and edges

Eg:



this is not a graph, its a particular drawing of a graph



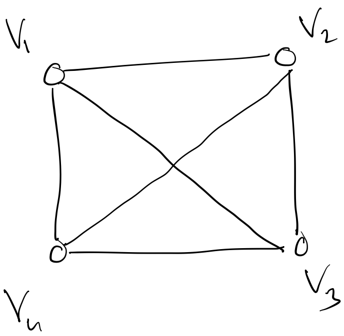
$$V = \{V_1, V_2, V_3, V_4\}$$

$$E = \{\{V_1,V_2\}, \{V_2,V_3\}, \{V_3,V_4\}, \{V_4,V_1\}\}$$

this is a graph

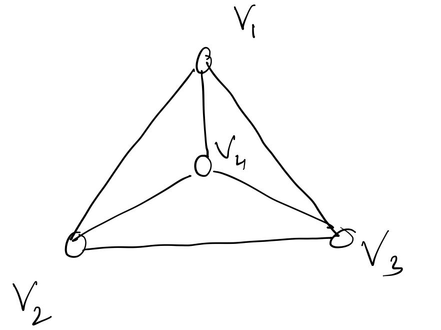
this is just a representation of a graph

K_4



this is not a planar graph, since there is an intersection

K_4



This is a planar graph

Adjacency Matrix

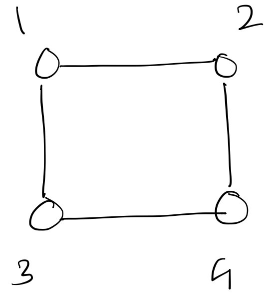
$A[i][j] = 1$ if i & j are adjacent

$= 0$ o.w

[Simple graph : no self loop, no multiple edges b/w two vertices]

We are dealing with simple graphs.

Eg:



$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

Space : $O(|V|^2)$

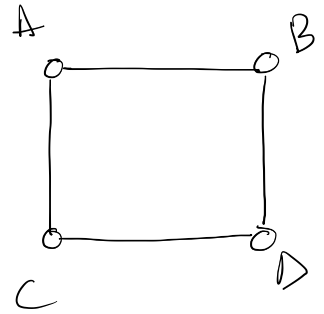
Check for an edge $O(1)$

traverse all neighbours of a particular vertex V : $O(|V|)$ [look at row or column]

The above matrix is good when graph is dense but it takes too much space.

Adjacency List

List of Adj vertices



A : B, C

B : A, D

C : A, D

D : B, C

Space : $O(|V| + |E|)$

Traverse neighbours : deg (v) (size of list for v)

Check if (u,v) exists : deg (u) (size of list for u)

Edge List

space : $O(|E|)$

(u,v) exists : $O(|E|)$.

traverse neighbours : $O(|E|)$

$\longrightarrow$ Dense graph : Adjacency Matrix

Sparse graph : Adjacency list